File

Name: R4-vendor-C-unidirectional
Description: KISSsoft example
Changed by: AAA on: 31.10.2026 at: 03:42:51

1 Overview

Calculation method: DIN 6892-C:2012

Label		DIN 6885.1:1968 Default
Key width (mm)	[b]	32.00
Key height (mm)	[h]	18.00
Shaft diameter (mm)	[d]	120.00
Nominal torque (Nm)	[Tn]	4000.00
Application factor	[KA]	1.50
equivalent torque (Nm)	[Teq]	6000.00
Maximum torque (Nm)	[Tmax]	15000.00
Minimal frictional torque for interference fit (Nm)	[TRmin]	0.00
Number of load peaks	[NL]	10000
Number of keys	[i]	1
Load factor	[phi]	1.00
equivalent circumferential stress (N)	[Feq]	100000.00
Maximal circumferential force (N)	[Fmax]	250000.00
Contact coefficient for equivalent surface pressure	[Kveq]	1.00
Contact coefficient for maximal surface pressure	[Kvmax]	1.00
Friction factor for equivalent torsional moment	[KReq]	1.000
Friction factor for maximum peak torsional moment	[KRmax]	1.000
Shaft groove depth (minimum value) (mm)	[t1]	11.00
Supporting key length (mm)	[ltr]	93.00
Supporting key height (mm)	[h-t1]	7.00
Equivalent effective surface pressure (N/mm²)	[peq]	153.61
Maximum effective surface pressure (N/mm²)	[pmax]	384.02

2 Shaft

Material, own Input		1 C 60 N
Type		Through hardened steel
Treatment		unalloyed/through hardened
Tensile strength (component's raw diameter) (N/mm²)	[Rm]	600.00
Yield point (component's raw diameter) (N/mm²)	[Rp]	310.00
Component raw diameter (mm)	[deff]	130
Permissible surface pressure (N/mm²)	[pzul]	279.00
Load peak frequency coefficient	[fL]	1.40
Safety factor, equivalent load (fw pzul/peq)	[SFeq]	1.82
Safety, maximum load (fL pzul/pmax)	[SFmax]	1.02
Minimal safety	[SF]	1.02

3 Hub

Material, own Input		EN GJL-250

Type		GJL (Cast iron flake graphite)
Treatment		untreated
Tensile strength (wall thickness) (N/mm²)	[Rm]	200.00
Yield point (wall thickness) (N/mm²)	[Rp]	130.00
The hub material strength values are defined using the wall thickness as the raw diameter.		
Component raw diameter (wall thickness) (mm)	[deff]	45
Permissible surface pressure (N/mm²)	[pzul]	180.00
Load peak frequency coefficient	[fL]	1.20
Safety factor, equivalent load (fw pzul/peq)	[SFeq]	1.17
Safety, maximum load (fL pzul/pmax)	[SFmax]	0.56
Minimal safety	[SF]	0.56

4 Key

Material, own Input		C45 K			
Type		Through hardened steel			
Treatment		unalloyed/through hardened			
Tensile strength (N/mm²)	[Rm]	680.00			
Diameter range (mm)	[d]	(19.0	-	19.0	)
Yield point (N/mm²)	[Rp]	430.00			
Diameter range (mm)	[d]	(19.0	-	19.0	)
Permissible surface pressure (N/mm²)	[pzul]	387.00			
Load peak frequency coefficient	[fL]	1.40			
Safety factor, equivalent load (fw pzul/peq)	[SFeq]	2.52			
Safety, maximum load (fL pzul/pmax)	[SFmax]	1.41			
Minimal safety	[SF]	1.41			

Remarks:
Safety = minimum (pzul/peq, fL pzul/pmax)
Condition according to DIN 6892 Safety >= 1.0
Supporting key height: h - t1

End of report (lines: 85)